

PROJECT CHARTER

Enterprise Manufacturing Data Platform (EMDP)

1. Project Purpose

ABC Manufacturing Ltd. operates five smart factories generating large volumes of data across production, quality, ERP, warehouse, inventory, maintenance, and IoT systems. This data is currently stored in separate silos, limiting operational visibility. The Enterprise Manufacturing Data Platform (EMDP) project will design and implement an end-to-end enterprise data platform to unify this data and enable production intelligence.

2. Project Objectives

- Collect manufacturing data from multiple heterogeneous sources.
- Cleanse and validate incoming data.
- Store curated datasets in an enterprise PostgreSQL warehouse.
- Maintain metadata and data lineage.
- Support analytics and reporting.
- Use Git for collaborative version control.
- Produce documentation expected in an enterprise project.

3. Project Scope

The project covers ingestion, staging, profiling, cleansing, warehousing, governance, and reporting of manufacturing data from all five factories. The solution will be delivered progressively over four Agile Scrum sprints using Pentaho, Python, PostgreSQL, Git/GitHub, and Power BI.

4. High-Level Timeline

Sprint	Focus Area	Key Modules
Sprint 0	Project Initiation & Architecture	Project planning and architecture
Sprint 1	Data Discovery & Ingestion	Data Types & Formats; Data Ingestion
Sprint 2	Data Profiling & DB / Data Warehouse	Data Profiling (Python); Storage & Retrieval (PostgreSQL)
Sprint 3	Data Governance, Lineage & Analytics	Data Lineage

5. Success Criteria

- A working solution is demonstrated at the end of every sprint.
- Project documentation is maintained continuously.
- A production-ready enterprise data platform is delivered progressively.
- All Definition of Done (DoD) criteria are met for every sprint.

6. Risks & Assumptions

6.1 Assumptions

- Representative sample data will be available for each source system.
- Required tools including Pentaho, PostgreSQL, Power BI, Git, and Python are available and accessible.

6.2 Risks

- Data quality issues in source systems may extend cleansing effort in later sprints.
- Scope creep may occur if additional source systems are identified during the project.

7. Project Governance

The project follows Agile Scrum. Work is organized into four sprints, with a working solution demonstrated at the end of each sprint. Project documentation is maintained continuously, and deliverables are reviewed as part of sprint governance.